

Started on Tuesday, 12 May 2026, 3:39 PM

State Finished

Completed on Tuesday, 12 May 2026, 3:43 PM

Time taken 3 mins 55 secs

Grade 14.00 out of 15.00 (93.33%)

Question 1 | Correct Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- ☐ a. Bubble search
- ☐ b. Merge search
- ☐ c. Quick search
- ☒ d. Linear search ✓

Question 2 | Correct Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- ☒ a. Checking each element sequentially ✓
- ☐ b. Dividing the list in half repeatedly
- ☐ c. Skipping every second element
- ☐ d. Sorting the list before searching

Question 3 | Correct Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- ☐ a. It compares elements sequentially
- ☐ b. It always starts from the beginning of the list
- ☒ c. It can be applied only if the list is sorted ✓
- ☐ d. It works on unsorted lists

Question 4 | Correct Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- ☐ a. When the item is not the array at all
- ☒ b. When the item is somewhere in the middle of the array ✓
- ☐ c. When the item is the last element in the array
- ☐ d. Item is the last element in the array or item is not there at all

Question 5 | Incorrect Mark 0.00 out of 1.00

What is the first step in binary search?

- ☐ a. Compare the target element with the middle element in the list
- ☐ b. Divide the list into two equal parts
- ☐ c. Compare the target element with the first element in the list
- ☒ d. Sort the list ✗

Question 6 | Correct Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n ?

- ☐ a. $n/2$
- ☐ b. $\log n$
- ☐ c. $n-1$
- ☒ d. n ✓

Question 7 | Correct Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n ?

- ☐ a. $n/2$
- ☒ b. n ✓
- ☐ c. $\log n$
- ☐ d. 1

Question 8 | Correct Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- ☐ a. The list is sorted
- ☐ b. The search continues in the left sublist
- ☐ c. The search continues in the right sublist
- ☒ d. The search ends successfully ✓

Question 9 | Correct Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- ☐ a. In the middle of the list
- ☒ b. In the right sublist ✓
- ☐ c. At the beginning of the list
- ☐ d. In the left sublist

Question 10 | Correct Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- ☐ a. Discovering
- ☐ b. Mining
- ☐ c. Finding
- ☒ d. Searching ✓

Question 11 | Correct Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- ☐ a. $O(n)$
- ☐ b. $O(n \log n)$
- ☐ c. $O(1)$
- ☒ d. $O(\log n)$ ✓

Question 12 | Correct Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- ☒ a. Linear search ✓
- ☐ b. Hashing
- ☐ c. Binary search
- ☐ d. Jump search

Question 13 | Correct Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- ☒ a. The search is considered unsuccessful ✓
- ☐ b. The last element is returned
- ☐ c. An error is raised
- ☐ d. The first element is returned

Question 14 | Correct Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- ☐ a. Binary search can find multiple instances of the target element
- ☐ b. Binary search does not require dividing the list
- ☒ c. Binary search has a lower time complexity on large, sorted lists ✓
- ☐ d. Binary search works on unsorted lists

Question 15 | Correct Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- ☐ a. Hash search
- ☐ b. Binary search
- ☒ c. Linear search ✓
- ☐ d. Both (1) & (3)